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Modeling CO₂ Leak and Dispersion from Wells and Surface Facilities: A Study on Enhancing Predictive Accuracy Using Machine Learning Models

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Abstract

Accurate modeling of CO₂ leak and dispersion is critical for effective emergency planning in carbon capture and storage operations. This study develops a machine learning-based model to estimate maximum downwind dispersion distances resulting from loss of primary containment events. A total of 13,464 PHAST simulations were conducted by systematically varying four key parameters: operating temperature (0–100 °C), pressure (1–250 bar), leak diameter (1–12 inches), and leak orientation (horizontal and impinging downward at 45°). For each case, the worst-case dispersion distances at 30,000 ppm and 40,000 ppm thresholds were extracted under two atmospheric stability classes (F and D) with corresponding surface wind speeds (1.5 m.s⁻¹ and 5 m.s⁻¹) respectively. Multiple machine learning models were evaluated, including Support Vector Regression, K-Nearest Neighbors, Decision Trees, Multi-Parametric Regression, and Artificial Neural Networks. Among these, the ANN model exhibited the highest performance with R² values exceeding 0.999 and root mean square error below 10 m on the test set. Sensitivity analyses confirmed the dominant influence of pressure and leak size on dispersion outcomes. Impinging leaks produced consistently longer dispersion distances due to the reduced release velocity following ground impact, which limits air entrainment into the cloud. Extensive diagnostic analyses, including residual, cumulative error, and leverage evaluations, confirmed the model's robustness and generalizability. The best performing machine learning model was integrated into an interactive computer application, enabling rapid hazard assessment, and offering a significant reduction in computational cost compared to Computational Fluid Dynamics (CFD) methods. This work provides a scalable framework for enhancing emergency response and process safety in CO₂-intensive industrial environments.

Keywords: CO₂ Leak and Dispersion, PHAST Simulation, Machine Learning, Emergency Response, CO₂ Well Integrity

Introduction

Anthropogenic greenhouse gas (GHG) emissions have driven a measurable rise in global temperatures, with projections estimating an increase of up to 5.9 °F by 2100 versus pre-industrial levels if current emission trajectories persist (Boehm and Schumer, 2023). This global warming poses critical risks to ecosystems, public health, and infrastructure. Carbon capture, utilization, and storage (CCUS) has emerged as a vital mitigation strategy to limit atmospheric CO₂ accumulation. To meet net-zero targets, CO₂ capture must expand to at least 4.2 gigatons per annum by 2050; a 120-fold increase from current capacity (Koytsoumpa et al., 2018; Yan and Zhang, 2019). Despite decarbonization efforts, CCUS introduces safety concerns, particularly regarding unintended CO₂ leakage. As a dense, odorless, and colorless asphyxiant, CO₂ can pose significant hazards even at moderate concentrations. Exposure levels at 40,000 ppm are classified by Occupational Safety and Health Administration (OSHA) as immediately dangerous to life and health (IDLH) and exposure levels at 30,000 ppm are classified by American Conference of Governmental Industrial Hygienists (ACGIH) as threshold limit value – short-term exposure limit (TLV-STEL) for a 15-minute period necessitating CCUS robust risk assessment frameworks to prevent any health risks to the public and operators near such CCUS facilities. Subsurface injection wells, integral to CCUS operations, may act as potential leak pathways if their integrity is compromised due to poor completion, long-term degradation, or operational failure (Aines et al., 2009; Gasda et al., 2004; Lewicki et al., 2007; Marbun et al., 2019). Previous field observations and incident reports from sites such as Crystal Geyser and the Travale geothermal field highlight the multifaceted risks of blowouts and surface leaks during CO₂ handling operations (Marbun et al., 2019; Skinner, 2003).

Accidental releases from wells, whether due to structural degradation, operational errors, or legacy abandonment, present complex dispersion behaviors, particularly when leak orientation interacts with surrounding topography. The orientation of a leak plays a critical role in governing CO₂ plume dynamics. One of the hazardous leak configurations arises when a high-pressure leak impinges downward at a 45° angle toward the ground. This configuration attenuates the initial release velocity as the cloud's momentum is absorbed upon impingement with the ground. The resulting reduction in velocity limits turbulence and ambient air entrainment, allowing the cloud to retain higher density for a longer duration during downwind transport. Consequently, this promotes greater lateral dispersion of hazardous gas concentrations. Accurately predicting such dispersion behavior is essential for delineating exclusion zones and informing emergency response protocols. The risk of unintended CO₂ releases during injection or workover operations remains a growing concern, particularly in mature fields where infrastructure has aged under decades of fluid exposure. The high-pressure environment and reactive nature of supercritical CO₂ can exacerbate the degradation of well materials, increasing the likelihood of mechanical failure or blowouts. These risks are especially pronounced during workover or re-completion activities, where pressure cycling and well interventions may compromise wellbore and wellhead integrity. In this context, the capacity to reliably model and predict the dispersion of CO₂ plumes becomes essential; not only for safeguarding personnel and infrastructure but also for guiding rapid emergency response protocols. Understanding how CO₂ disperses upon release, especially under complex leak geometries, is therefore a foundational requirement for safe and effective deployment of carbon sequestration technologies.

Historically, gas dispersion has been modeled using empirical approaches such as Gaussian plume models, which, while fast, lack the resolution to account for complex release geometries, terrain effects, and thermophysical interactions. Computational Fluid Dynamics (CFD) models offer improved fidelity but remain computationally intensive and difficult to generalize across scenarios (Liu et al., 2020; Shen et al., 2020; Zhang et al., 2019). Integrated models like PHAST have gained wide acceptance in process safety due to their capacity to simulate toxic and flammable gas dispersion with validated physical correlations. However, PHAST's utility in real-time applications is constrained by its deterministic nature and simulation runtime. The integration of machine learning (ML) with physics-based simulators has

recently been explored to bridge this gap. Models trained on PHAST outputs can generalize complex dispersion behavior under a wide range of operating conditions with minimal computational overhead. Prior studies have shown the applicability of gradient boosting, deep neural networks, and support vector machines to reproduce dispersion distances with high fidelity (Abdalla et al., 2025; Fan et al., 2024; Jiao et al., 2021; Xiao et al., 2022). This approach enables the rapid prediction of exclusion zones while accounting for nonlinear interactions among input features such as pressure, temperature, leak diameter, and ambient weather conditions.

Building upon our previously published study that established a robust machine learning framework for predicting CO₂ dispersion from horizontal leaks in injection wells and surface facilities based on PHAST-generated simulation data (Abdalla et al., 2025), the present work extends the methodology to address a critical yet understudied scenario: impinging leaks at a 45° downward angle toward the ground. To capture this effect, a fourth input variable, Leak Orientation, is introduced into the model, numerically coded to distinguish horizontal releases from impinging downward leaks. This enhancement enables a comparative assessment of dispersion behavior under varied geometrical release configurations, supporting more conservative and realistic exclusion zone predictions. The updated model is trained on an expanded PHAST simulation dataset encompassing both orientations, thereby reinforcing the utility of machine learning for real-time consequence prediction in CCUS safety assessments and enables real-time deployment in emergency planning scenarios for CCUS infrastructure.

While previous studies have demonstrated the ability of machine learning models to predict CO₂ dispersion patterns from physics-based simulations, the present work introduces several key advancements. First, the input variables are expanded to include leak orientation as a fourth independent variable, capturing the distinction between horizontal releases and ground-impinging leaks at 45°, a geometry with practical relevance in wellhead failures and surface equipment leaks. Second, the simulation data curation spans a wide operational envelope, temperature (0–100 °C), pressure (1–250 bar), and leak size (1–12 inches), ensuring that the model is applicable to a broad range of potential real-world leak scenarios. Third, the model training pipeline includes five machine learning algorithms, each subjected to hyperparameter optimization and validated via cross-validation to identify the best-performing model. Fourth, this study enhances applicability of dispersion modeling and real-time emergency response by embedding the best trained machine learning model into a lightweight, Python-based computer application that enables real-time prediction of exclusion zones, supporting immediate risk assessment during emergency events. Together, these innovations deliver a scalable, fast, and broadly generalizable prediction tool that extends beyond earlier Machine Learning–PHAST integrations in both scope and performance.

Methodology

Dispersion Simulation

To simulate the physical behavior of CO₂ following a high-pressure release, this study employs PHAST, an industry-standard consequence modeling software developed by DNV. PHAST integrates multiple coupled sub-models to resolve complex gas dispersion phenomena including depressurization, jet release, entrainment, vaporization, and atmospheric mixing. These models are built upon conservation equations of mass, momentum, and energy, and are capable of accounting for buoyancy effects, turbulence, and phase changes, all of which are critical for modeling the dynamic behavior of CO₂ clouds. This makes PHAST a highly suitable tool for generating high-fidelity dispersion data under varied leak scenarios. The tool's ability to simulate the evolution of CO₂ clouds under different meteorological and release conditions allows for the generation of robust training data for subsequent machine learning workflows. (Jiao et al., 2021; Zhong et al., 2024).

Data Point Generation from Simulations

In this study, a comprehensive dataset was generated by executing a large number of PHAST simulations, each corresponding to a unique combination of input parameters. Four independent variables were considered: operating temperature (0–100 °C), pressure (1–250 bar), leak size (1–12 inches), and a new parameter introduced in this study; leak orientation. The leak orientation variable captures the effect of release geometry, where 0 corresponds to horizontal leaks and 1 denotes impinging downward leaks at 45°. This additional degree of realism aims to better reflect worst-case accidental scenarios, where a ground-directed CO₂ release can lead to accumulation and extended exclusion zones due to suppressed plume rise and reduced entrainment.

Other scenario parameters were held constant to ensure comparability across runs, including a fixed release height of 1 meter, consistent ambient temperature and pressure, and two atmospheric stability classes: Class D (neutral, with 5.0 m/s wind speed) and Class F (stable, with 1.5 m/s wind speed). These classes reflect globally relevant meteorological conditions and represent typical daytime and highly stable limited atmospheric mixing nighttime environments, respectively (Epa and of Emergency Management, 1987). For each simulation, PHAST computes the maximum downwind distance at which CO₂ concentrations reach health-critical thresholds of 30,000 ppm (TLV-STEL) and 40,000 ppm (IDLH). These thresholds are selected to evaluate the extent of hazardous zones requiring emergency response planning.

Data Preprocessing and Feature Engineering

Following running the PHAST simulation, the dataset is curated to ensure readiness for machine learning model development. Duplicate records and missing values are removed, and the data are split using a fixed random seed into a training set (10,771 samples) and a testing set (2,693 samples), preserving an 80:20 ratio. All continuous input variables are standardized using StandardScaler, which transforms features to have zero mean and unit variance. This is essential for ensuring stable and efficient training across various algorithms, particularly when dealing with variables of differing physical scales such as pressure and temperature. The inclusion of the new leak orientation feature is integrated into the model input structure, allowing the trained machine learning models to learn and generalize the influence of impinging leaks relative to horizontal ones. Exploratory data analysis, including correlation matrices and distribution plots, is performed to verify feature relevance and to rule out multicollinearity or data imbalance. The final dataset thus forms a physically informed, statistically sound foundation for developing predictive models that are both accurate and generalizable across diverse leak conditions.

Machine Learning Model Development

Model Selection and Rationale. To accurately capture the complex and nonlinear interactions between CO₂ dispersion distances and the expanded set of input variables, including temperature, pressure, leak size, and the newly introduced leak orientation, this study adopts a comparative machine learning approach. A diverse suite of algorithms is selected, each representing a distinct modeling philosophy, ranging from kernel-based learning to ensemble techniques. The selected models include:

- Support Vector Regression (SVR): it extends the principles of support vector machines to regression tasks by introducing an ϵ -insensitive loss function. It maps input features into a higher-dimensional space (via kernel functions) to find a function that fits the data within a specified margin. This margin-based approach often yields robust performance on datasets with moderate noise and can handle high-dimensional feature spaces effectively. However, SVR requires careful tuning of parameters such as the kernel type, regularization constant C , and the ϵ margin to avoid overfitting or underfitting (Abdalla et al., 2025).

- K-Nearest Neighbors (KNN): A non-parametric approach that estimates outcomes by averaging the responses of the closest points in the feature space (Jiao et al., 2020). Its simplicity and minimal assumptions about data distribution make it a valuable benchmark for comparative analysis, though performance can degrade with higher-dimensional data or large datasets (Abdalla et al., 2025).
- Artificial Neural Networks (ANN): Featuring multi-layer perceptron architectures, ANNs are adept at modeling intricate nonlinear relationships. Their ability to learn hierarchical feature representations is particularly useful for capturing the subtle effects of varying temperature, pressure, and leakage geometry on dispersion behavior. However, ANNs require larger datasets and careful hyperparameter tuning (e.g., number of layers, learning rate) to avoid overfitting (Gomaa et al., 2024).
- Multi-Parametric Regression (MPR): This model extends traditional linear regression by incorporating polynomial and interaction terms. It offers a balance between interpretability and the ability to capture moderate nonlinearities, although it may struggle with highly complex data patterns.
- Decision Trees (DT): By partitioning the input space into discrete regions based on feature thresholds, decision trees can effectively model nonlinear interactions. Their inherent interpretability further aids in understanding the contribution of individual parameters to dispersion outcomes. DTs are easy to interpret and relatively fast to train but can exhibit high variance and overfitting if not regularized (e.g., through pruning or minimum samples per leaf). The rationale behind this multi-model strategy is to systematically compare diverse modeling paradigms in terms of predictive accuracy and computational efficiency (Abdalla et al., 2025).

Training Strategy and Evaluation Metrics. The dataset is partitioned into an 80% training set and a 20% testing set to ensure robust model development. For each model, training is performed on the 80% subset, and model performance is validated using 5-fold cross-validation with a fixed random seed for reproducibility. Model performance is assessed using three metrics, as follows:

- Root Mean Squared Error (RMSE): Measures the square root of the average squared differences between predicted and observed values, emphasizing larger errors.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad \text{Equation 1}$$

(Abdalla et al., 2025)

- Mean Absolute Error (MAE): Computes the average absolute difference between predicted and observed values, offering a more interpretable view of model precision.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad \text{Equation 2}$$

(Abdalla et al., 2025),

- Coefficient of Determination (R^2): Indicates the proportion of variance in the observed data explained by the model, with a value of 1 representing perfect prediction.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad \text{Equation 3}$$

(Abdalla et al., 2025)

This multi-metric evaluation ensures that each model is optimized not only for minimizing absolute and squared prediction errors but also for accurately capturing the variance in the CO₂ dispersion distance across different leak scenarios.

Hyperparameter Tuning and Optimization. Hyperparameter optimization is critical to tailoring each model's performance to the dispersion prediction task. For each model, we employ a randomized search strategy using RandomizedSearchCV to efficiently explore the hyperparameter space. For example, in the KNN model, parameters such as the number of neighbors, search algorithm ('auto', 'ball_tree', 'kd_tree', 'brute'), and leaf size are tuned. This optimization process is integrated with the cross-validation framework ensuring that lower prediction errors are prioritized. Once the optimal hyperparameters are identified, the model is retrained on the full training set. This systematic hyperparameter tuning approach is applied consistently across all models (ANN, MPR, SVR, KNN, and DT), ensuring that each model is optimized for predictive performance while maintaining computational efficiency suitable for real-time applications (Abdalla et al., 2025).

Results and Discussion

Dispersion Profiles

The PHAST simulation results reveal the spatial extent of hazardous CO₂ dispersion for both horizontal and impinging leak orientations, across the two concentration thresholds of 30,000 ppm (TLV-Ceiling) and 40,000 ppm (IDLH). For each simulation case, the maximum downwind dispersion distance was extracted from the two atmospheric conditions tested, thereby representing the worst-case scenario per input configuration. This resulting dataset. Table 1 provides summary statistics for these maximum downwind distances (in meters), showing a robust basis for evaluating how leak geometry influences ground-level dispersion risks.

Table 1—Descriptive statistics for maximum dispersion distances under two leak orientation scenarios.

LEAK ORIENTATION	THRESHOLD	MEAN	STANDARD DEVIATION	MIN	25 TH PERCENTILE	50 TH PERCENTILE	75 TH PERCENTILE	MAX
HORIZONTAL	30,000 ppm	371.82	298.35	2.71	111.71	292.75	585.58	1137.18
HORIZONTAL	40,000 ppm	281.24	232.69	2.09	79.54	215.84	445.79	890.57
IMPINGING DOWNWARD AT 45°	30,000 ppm	711.79	514.58	15.00	274.00	591.00	1090.00	2117.00
IMPINGING DOWNWARD AT 45°	40,000 ppm	522.60	375.27	10.00	203.75	437.50	793.00	1612.00

Statistical analysis of the maximum dispersion distances indicates a consistent and substantial enhancement in downwind reach for impinging leaks compared to horizontal releases. For instance, the mean dispersion distance for impinging leaks at 30,000 ppm reaches approximately 712 m, nearly double that of the horizontal case (372 m), while the corresponding 40,000 ppm threshold shows an increase from 281 m to 523 m. These differences are further reflected in the upper extremes, where downward impinging leaks extend as far as 2,117 m, compared to a maximum of 1,137 m for horizontal leaks at the 30,000 ppm threshold.

Figure 1 presents the distribution histograms of maximum downwind distances at both concentration thresholds (30,000 and 40,000 ppm) for the two leak orientations. The histograms illustrate right-skewed distributions for all cases, with impinging leaks showing broader tails extending beyond 2,000 m for the

30,000 ppm threshold and up to 1,600 m for 40,000 ppm. These elongated distributions suggest a higher likelihood of large exclusion zones when the leak jet is directed downward, consistent with suppressed plume buoyancy and stronger ground-level accumulation. In contrast, horizontal leaks, while still hazardous, exhibit tighter distributions with lower mean and maximum distances.

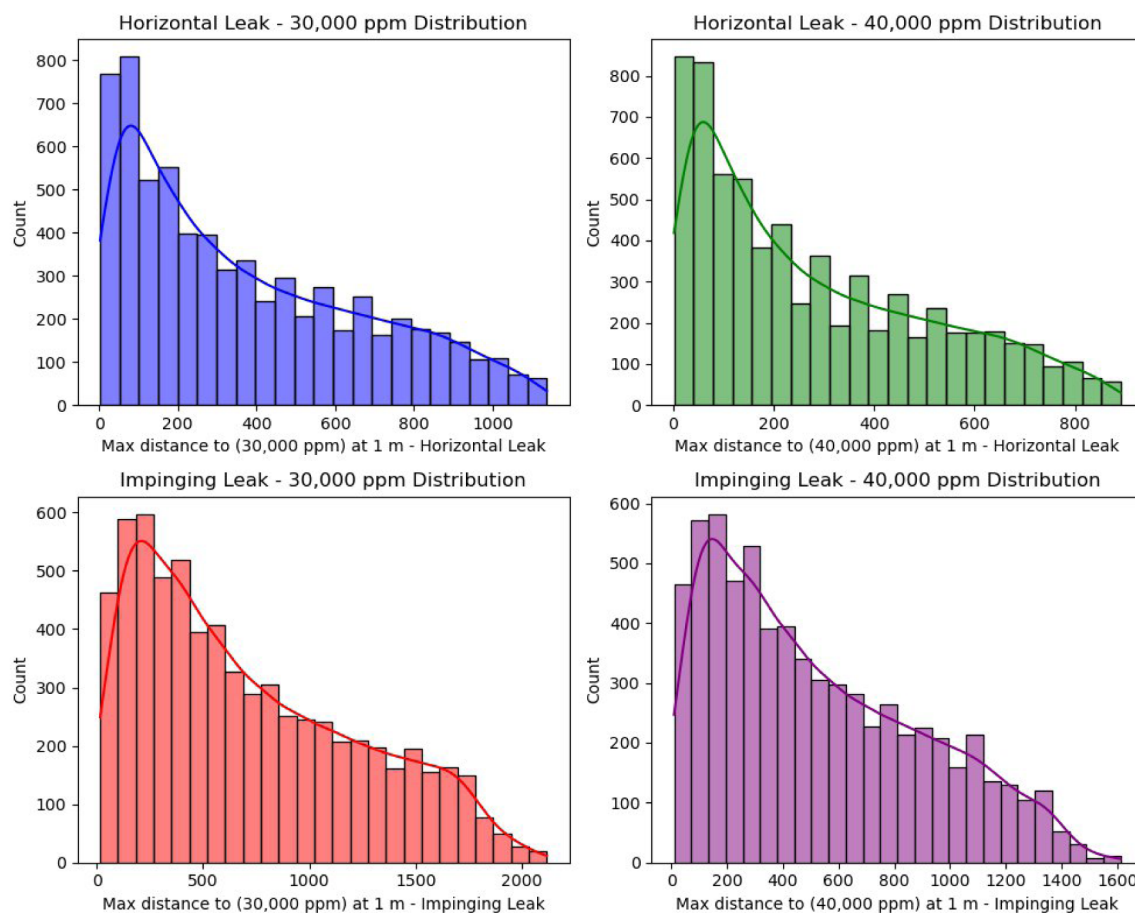


Figure 1—Histograms of the maximum downwind distances for CO₂ under two leak orientations (horizontal and impinging downward at 45°) and two concentration thresholds (30,000 ppm for STEL and 40,000 ppm for IDLH).

Figure S1 explores the relationships between temperature and dispersion distance. Across both leak orientations, higher temperatures are consistently associated with reduced dispersion ranges. This inverse relationship is attributed to the diminished density contrast between the released CO₂ and the surrounding atmosphere at elevated temperatures, which increases the kinetic energy and dissipation of the gas plume (Abdalla et al., 2025). Higher temperature CO₂ cloud possesses higher internal energy, which increases its tendency to expand and mix more rapidly with ambient air upon release, enhancing dilution near the source. This inverse trend is statistically supported by moderate negative correlation coefficients of approximately -0.33 to -0.36 , as shown in the correlation matrix (Figure 2). The trend is more pronounced under impinging downward leaks, where lower temperatures amplify the ground-level accumulation of the dense CO₂ cloud that has lower kinetic energy to disperse with air compared to the higher temperature cloud. Pressure effects are depicted in Figure S2, showing that increasing CO₂ pressure leads to longer downwind distances due to higher release rates. This monotonic trend is apparent across both thresholds and leak orientations. Notably, for impinging releases, the impact of pressure is amplified due to the plume's direct interaction with the ground, resulting in laterally spreading gas cloud over longer distances that results from high gas release rate coupled with lower velocity. This nonlinear but strong positive correlation between pressure and maximum downwind distance (correlation coefficient: $0.48 - 0.51$) is especially critical in impinging

leaks, where released gas directed toward the ground create dense gas pools that spread laterally along the surface. At 30,000 ppm concentration of CO₂, impinging releases at 250 barg exceed 2,000 m in some cases, underscoring the compounded effect of pressure and leak directionality on risk extent. Leak size, the most dominant factor, showed the strongest correlation with maximum dispersion distances among all input variables (correlation coefficient: 0.65–0.66). As displayed in Figure S3, larger leak diameters produced broader clouds due to increased volumetric flow rates. The increase is nearly linear for both leak orientations and concentration thresholds. For instance, horizontal releases through a 12-inch leak yielded maximum distances above 1,100 m at 30,000 ppm, while the impinging equivalent exceeded 2,000 m. These trends indicate that leak size plays a pivotal role in governing both the initial mass flux and momentum of the CO₂ jet, with larger diameters resulting in more concentrated and forceful releases. In impinging leak scenarios, the lower cloud velocity, due to interaction with the ground, enhances near-ground accumulation and extends the lateral dispersion footprint, reinforcing the importance of leak size as a critical parameter in both hazard assessment and model input prioritization.

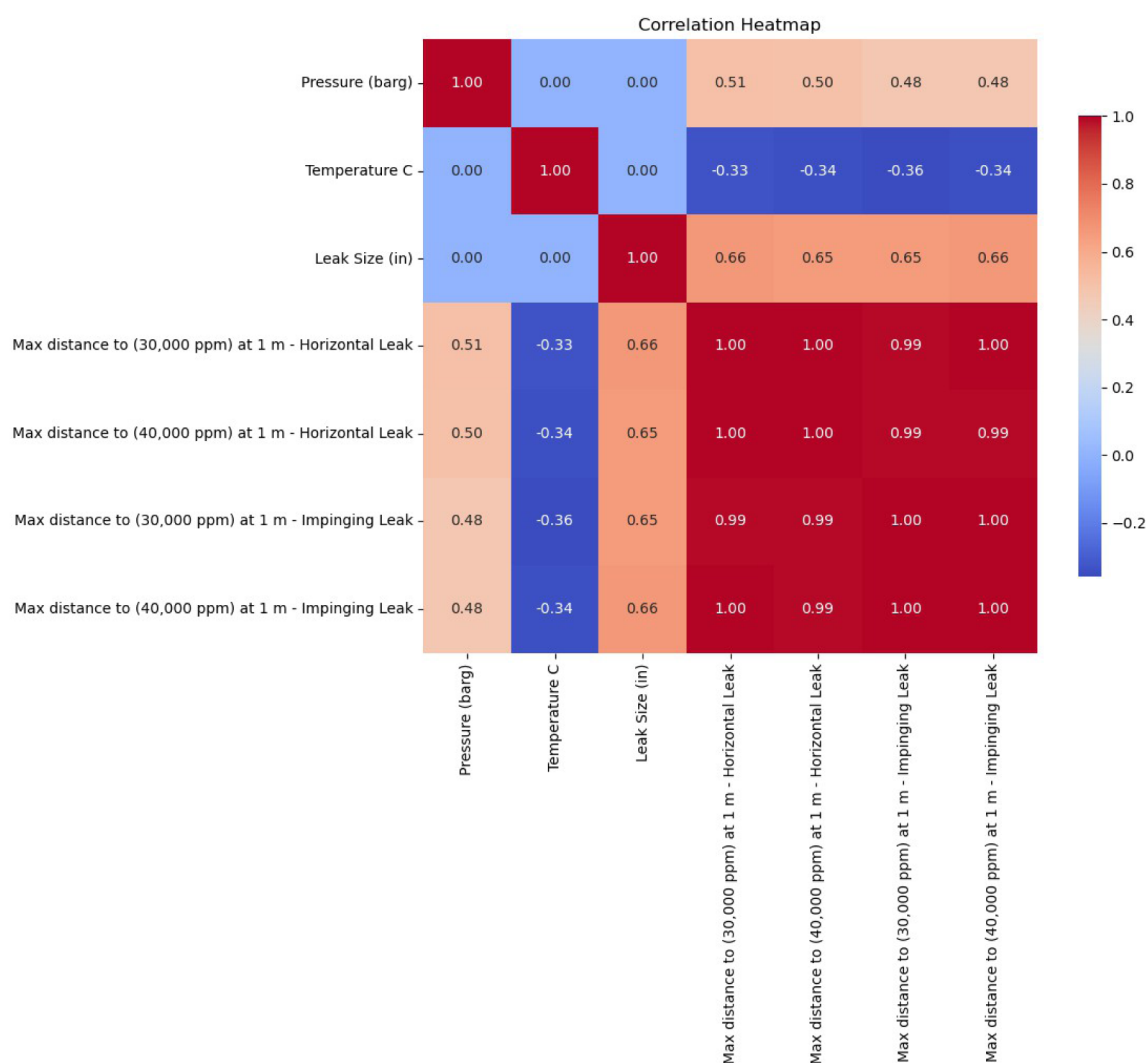


Figure 2—Correlation heatmap illustrating the relationships between the operational parameters and the maximum downwind distances.

These patterns are further illustrated in Figure 3, where three-dimensional scatter plots map the complex interplay between pressure, temperature, and leak size against the resulting dispersion distances. The plots reveal consistent trends across both horizontal and impinging orientations: larger leaks and higher

pressures result in more extensive downwind spread. In contrast, temperature demonstrates a weaker, inverse relationship with dispersion range, as previously discussed, likely due to its effect on plume buoyancy and gas density contrast with ambient air. Understanding the relative influence of each input variable is essential for effective model training. The high correlation of pressure and leak size with dispersion outcomes informs feature weighting strategies, ensuring that machine learning algorithms can effectively prioritize the variables that most significantly impact public and operational safety.

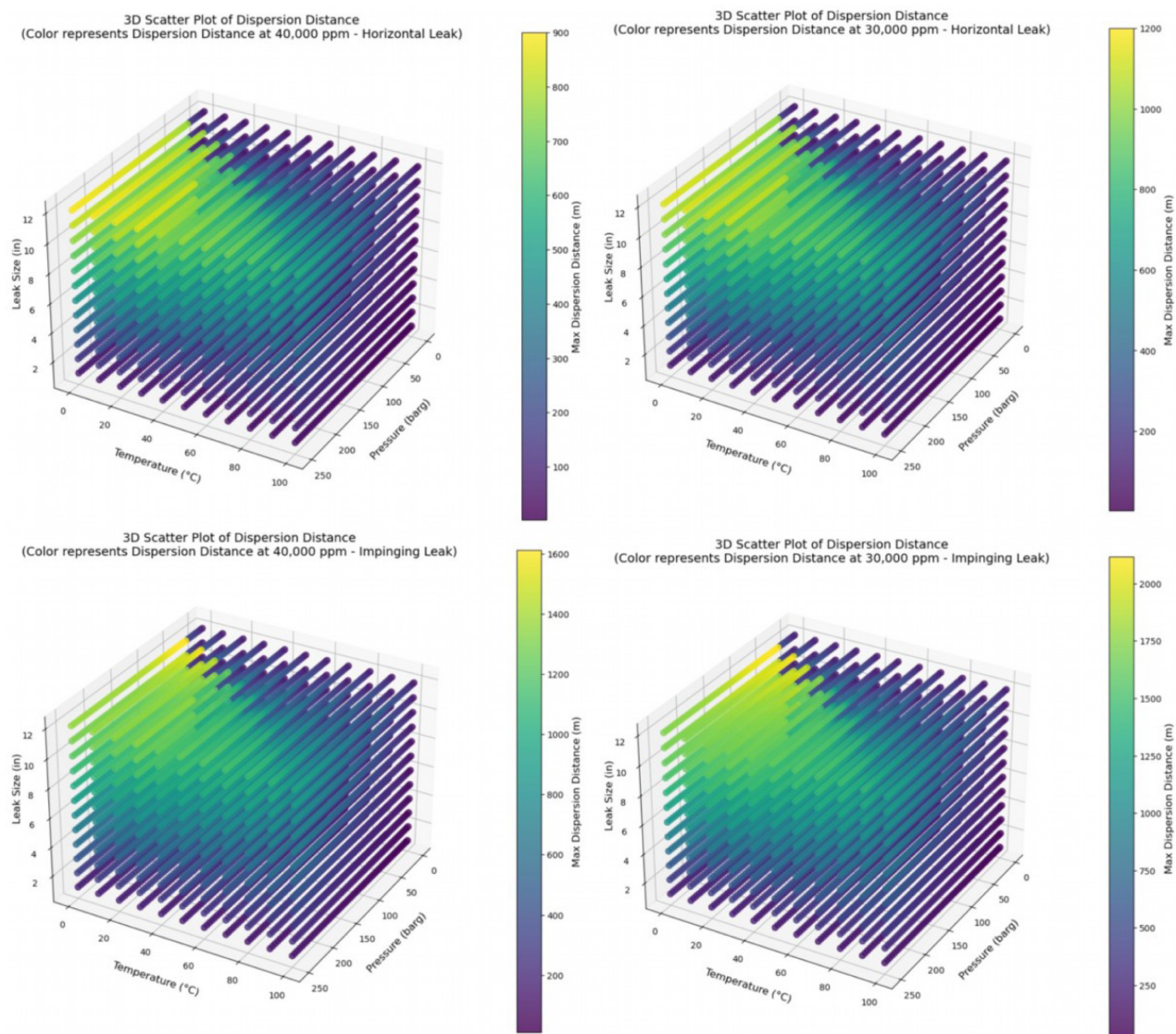


Figure 3—Three-dimensional scatter plots illustrating the joint influence of pressure, temperature, and leak size on CO₂ dispersion distances from horizontal and impinging downward leaks at STEL and IDLH thresholds.

Performance Evaluation

Five machine learning models; SVR, KNN, DT, ANN, and MPR, are compared using the generated PHAST simulation dataset. All models were trained using optimized hyperparameters, and the tuning outcomes revealed valuable insights into each algorithm's behavior. The SVR model achieved its best performance using a high regularization parameter ($C = 62$) with a radial basis function (RBF) kernel, indicating a need to balance model flexibility and generalization for capturing complex nonlinear dispersion relationships. For KNN, the optimal configuration involved 9 neighbors, the brute-force algorithm, and a leaf size of 70 with the Euclidean distance metric, highlighting the model's reliance on higher-resolution local patterns in

the data. The decision tree model performed best with a depth of 8 and 73 leaf nodes, indicating moderate model complexity was sufficient to learn key threshold-based rules. The ANN model showed the highest predictive accuracy when using 3 hidden layers with 20 neurons each and a learning rate of 0.001, supporting the benefit of deeper architectures with controlled learning rates in capturing nonlinearities. Lastly, the MPR model achieved optimal results with a 10th-order polynomial, which effectively fit the high-order interactions among pressure, temperature, and leak size. A summary of the hyperparameter search ranges and the final selected values for each model is presented in [Table 2](#).

Table 2—Hyperparameter tuning ranges and optimal values for each machine learning model used in this study

MACHINE LEARNING TECHNIQUE	HYPERPARAMETERS	RANGE TESTED	OPTIMIZED VALUE SELECTED
SVR	Regularization parameter; C	1 – 100	62
	Gamma	0.01 – 1.5	0.9376
	Kernel	linear, sigmoid, poly, rbf	rbf
	Degree	1 – 10	1
KNN	Number of Neighbors	3 – 20	9
	Algorithm	Auto, Ball Tree, KD Tree, Brute	Brute
	Leaf Size	10 – 100	70
	Distance Metric	Minkowski, Euclidean, Manhattan	Euclidean
DT	Maximum Depth	2 – 10	8
	Minimum Samples Split	2 – 10	6
	Maximum Leaf Nodes	2 – 100	73
ANN	Hidden Layers	2 – 4	3
	Neurons per Layer	20, 40, 60	20
	Learning Rate	1e-4, 1e-3, 3e-3	1e-3
MPR	Degree	1 – 10	10

Model Accuracy and Generalization. Model performance is assessed using RMSE, MAE, and R^2 across training and testing datasets, as summarized in [Table 3](#). The ANN achieves the lowest test RMSE (8.49 m), lowest test MAE (5 m) and highest test R^2 (0.9993) as shown in [Figure 4](#), indicating exceptional predictive accuracy and minimal overfitting. KNN ranks second in terms of test RMSE (33.82 m), test MAE (14.64 m) and test R^2 (0.9895). MPR, SVR, and DT follow, with DT showing the largest discrepancy between training and testing errors. Visual representations in [Figure 5](#) display predicted versus measured dispersion distances and model training/validation behavior, respectively. These plots further confirm ANN's superior fit and the robustness of the trained models under varying input conditions.

Table 3—Performance comparison of machine learning models on training and testing datasets.

MODEL	RMSE (TRAINING)	RMSE (TEST)	MAE (TRAIN)	MAE (TEST)	R ² (TRAIN)	R ² (TEST)
SVR	49.0162	48.4357	16.3892	16.6992	0.9787	0.9784
KNN	34.0603	33.8228	13.2597	14.6395	0.9897	0.9895
DT	68.1831	73.5660	49.5185	51.6766	0.9588	0.9502
ANN	7.7894	8.4935	4.7118	4.9953	0.9995	0.9993
MPR	39.5932	41.0815	18.3292	19.4355	0.9861	0.9845

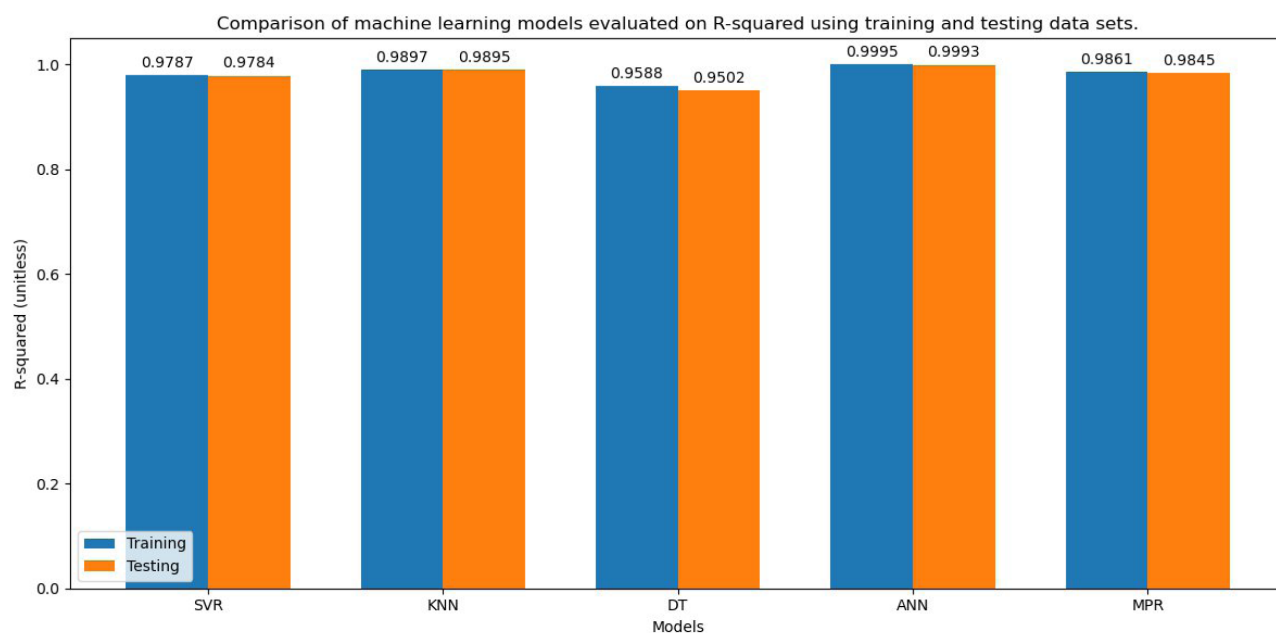


Figure 4—Comparison of machine learning models based on their R² values on both training and testing datasets at IDLH (40,000 ppm) threshold. Each pair of bars represents a single model (SVR, KNN, DT, ANN, and MPR), illustrating how effectively each model explains the variance in dispersion distance.

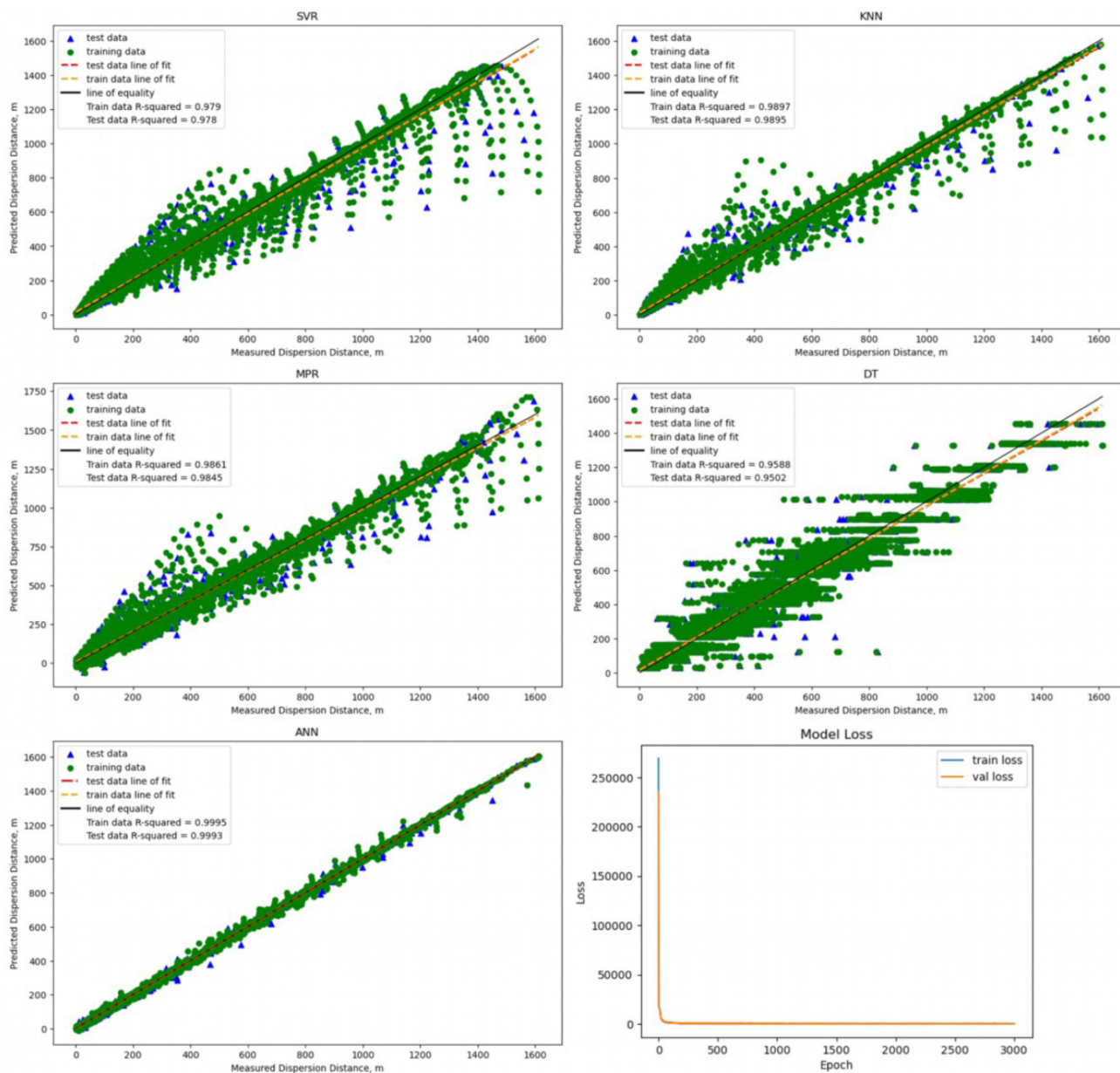


Figure 5—Predicted versus measured CO₂ dispersion distances at IDLH threshold for training and test data across multiple machine learning models.

Visual Examination of Model Outputs. Each scatter plot includes a line of fit for both the training and test data, as well as the line of equality ($y = x$). While KNN, MPR, and DT show a generally strong alignment between predictions and measurements, the ANN exhibits near-perfect alignment, with points clustering tightly around the line of equality. The ANN training and validation loss curves converge rapidly and remain low over thousands of epochs, confirming stable training and robust generalization to unseen data.

The SHAP (Shapley Additive exPlanations) in Figure 6 plot further reinforces the relationships observed in the scatter plots (Figures S1-S3), quantitatively demonstrating the influence of each input variable on the model's output. Leak size and pressure exhibit the highest SHAP values based on the best performing ANN model, confirming their dominant roles in driving CO₂ dispersion distances, consistent with their strong positive correlations and monotonic trends in the scatter plots. Temperature shows a more moderate but still notable effect, where lower temperatures (blue points) are generally associated with longer predicted dispersion distances, aligning with the physical behavior of denser CO₂ plumes under cooler conditions.

Importantly, leak orientation emerges as impactful categorical input variable, with impinging downward leaks (assigned zero value, as low feature value) consistently pushing the SHAP values toward greater dispersion distances compared to horizontal leaks (assigned one value, as high feature value), underscoring the added hazard posed by ground-directed releases.

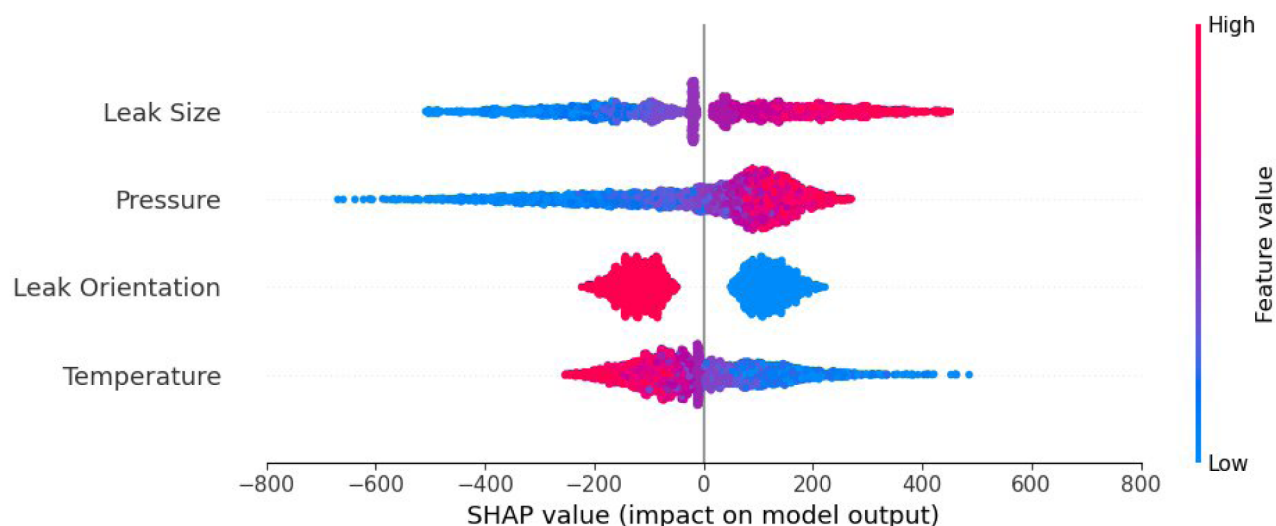


Figure 6—SHAP summary plot illustrating the impact of each input variable; leak size, pressure, leak orientation, and temperature, on the predicted CO₂ dispersion distance.

Error Distribution and Residuals. With a minimal gap between training and testing RMSE, the ANN's residuals are both relatively small in magnitude and uniformly distributed, indicating that the model captures most of the variance in the dataset compared to other models as indicated in Figures 7, 8. By contrast, the SVR and DT models exhibit wider spread of errors, suggesting that they struggle to capture the complexity of the dispersion relationships, especially at higher distance values. The choice of model is guided by a balance of predictive accuracy, generalization capacity, and computational feasibility for real-time applications. Despite the KNN model's strong performance ($R^2 \approx 0.9895$ on test data), the ANN outperforms it in every metric, achieving an R^2 of 0.9993 and an RMSE below 9 m. This high accuracy and minimal overfitting make the ANN particularly well-suited for real-time emergency response, where precise dispersion distance estimates are crucial for safety decisions. Given the wide range of temperatures, pressures, and leak sizes in the dataset, the ANN's ability to handle complex nonlinear interactions is a decisive advantage. Although ANNs can be computationally demanding, the dataset size (13,464 samples) remains manageable, and modern hardware or cloud-based solutions can efficiently deploy the model in real-time settings. Therefore, the ANN is selected as the primary predictive model for CO₂ dispersion, providing rapid, accurate distance estimates at both STEL and IDLH thresholds under diverse meteorological and leak orientation conditions.

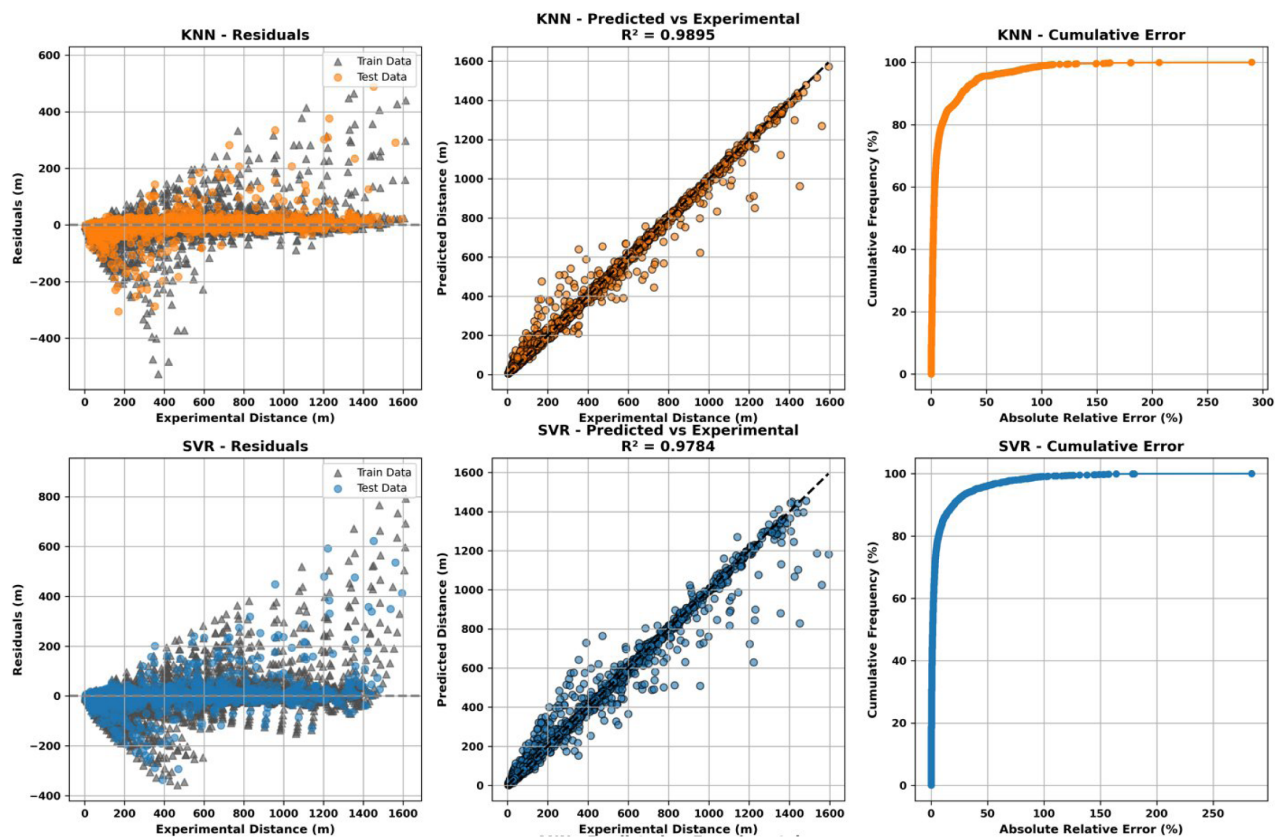


Figure 7—Diagnostic plots for the IDLH case (40,000 ppm) across multiple machine learning models. Top row corresponds to a KNN and bottom row corresponds to SVR, and the three columns depict: (1) residuals versus experimental distance, (2) predicted versus experimental distance (with R^2), and (3) cumulative frequency of the absolute relative error.

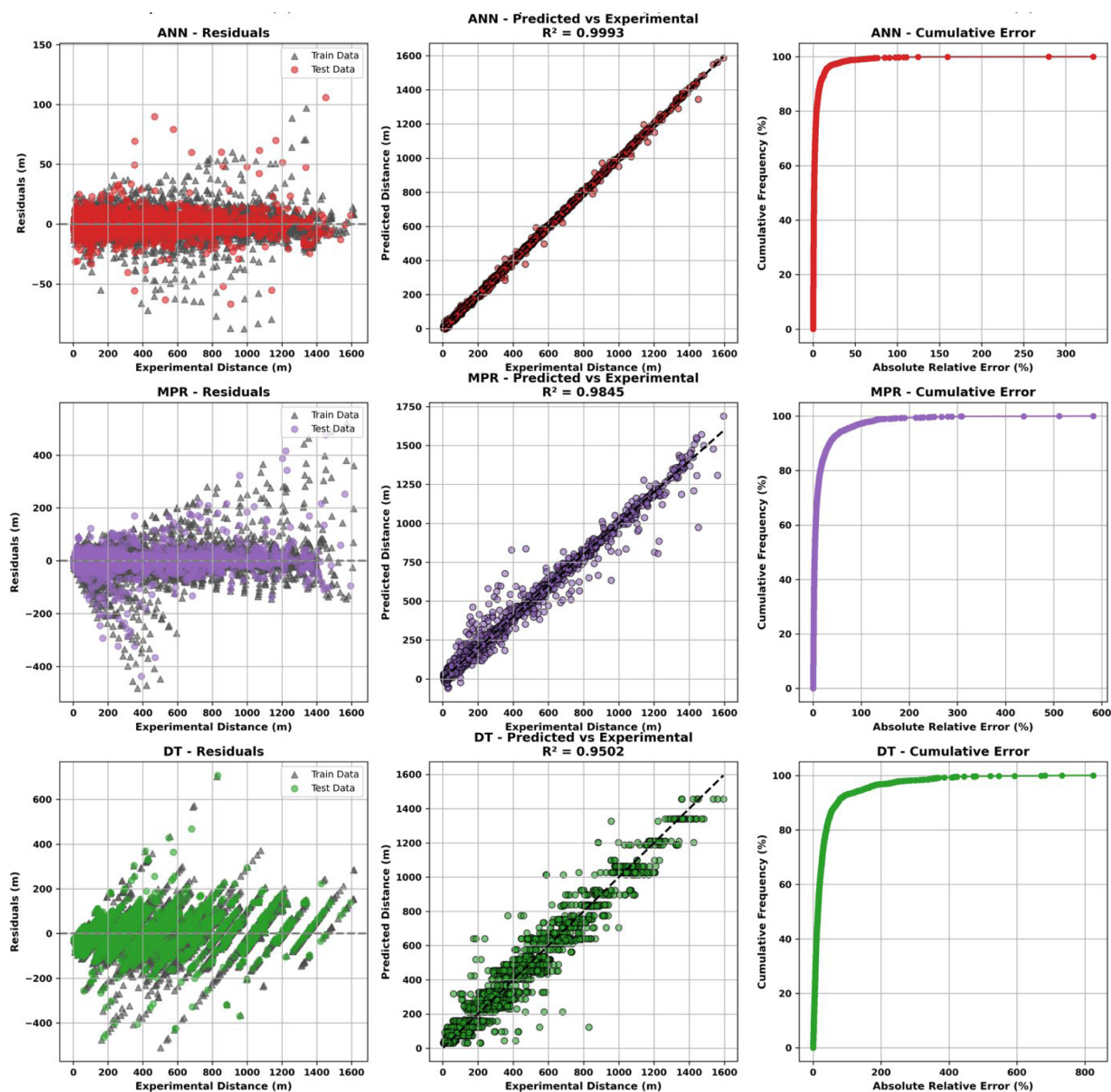


Figure 8—Diagnostic plots for the IDLH case (40,000 ppm) across multiple machine learning models.

Top row corresponds to a ANN, middle row corresponds to MPR, and bottom row corresponds to SVR, and the three columns depict: (1) residuals versus experimental distance, (2) predicted versus experimental distance (with R^2), and (3) cumulative frequency of the absolute relative error.

Practical Implications and Limitations. The high accuracy and rapid predictive capability of the selected ANN model offer significant practical benefits for emergency response and safety management in industrial environments. This level of precision is critical in defining safe evacuation zones and deploying emergency response measures swiftly, especially in facilities where CO₂ is stored or utilized under high-pressure conditions. By integrating these predictions into an interactive computer application, developed using python as shown in Figure 9, safety managers can quickly assess exclusion zones in real-time, ensuring that response plans are both proactive and accurately tailored to the specific conditions of a potential leak. Thus far, our methodology demonstrates a significant reduction in computational burden compared to conventional CFD simulations, as the predictive model leverages pre-simulated data from PHAST. This efficiency not only facilitates near-instantaneous predictions but also enables the continuous updating and retraining of the model as new simulation data becomes available or extended range of operation conditions.

Ultimately, the deployment of an ANN-based dispersion predictor promises to improve overall process safety, reduce downtime in emergency scenarios, and support the development of more robust, data-driven risk management strategies in the chemical and petroleum process industry.

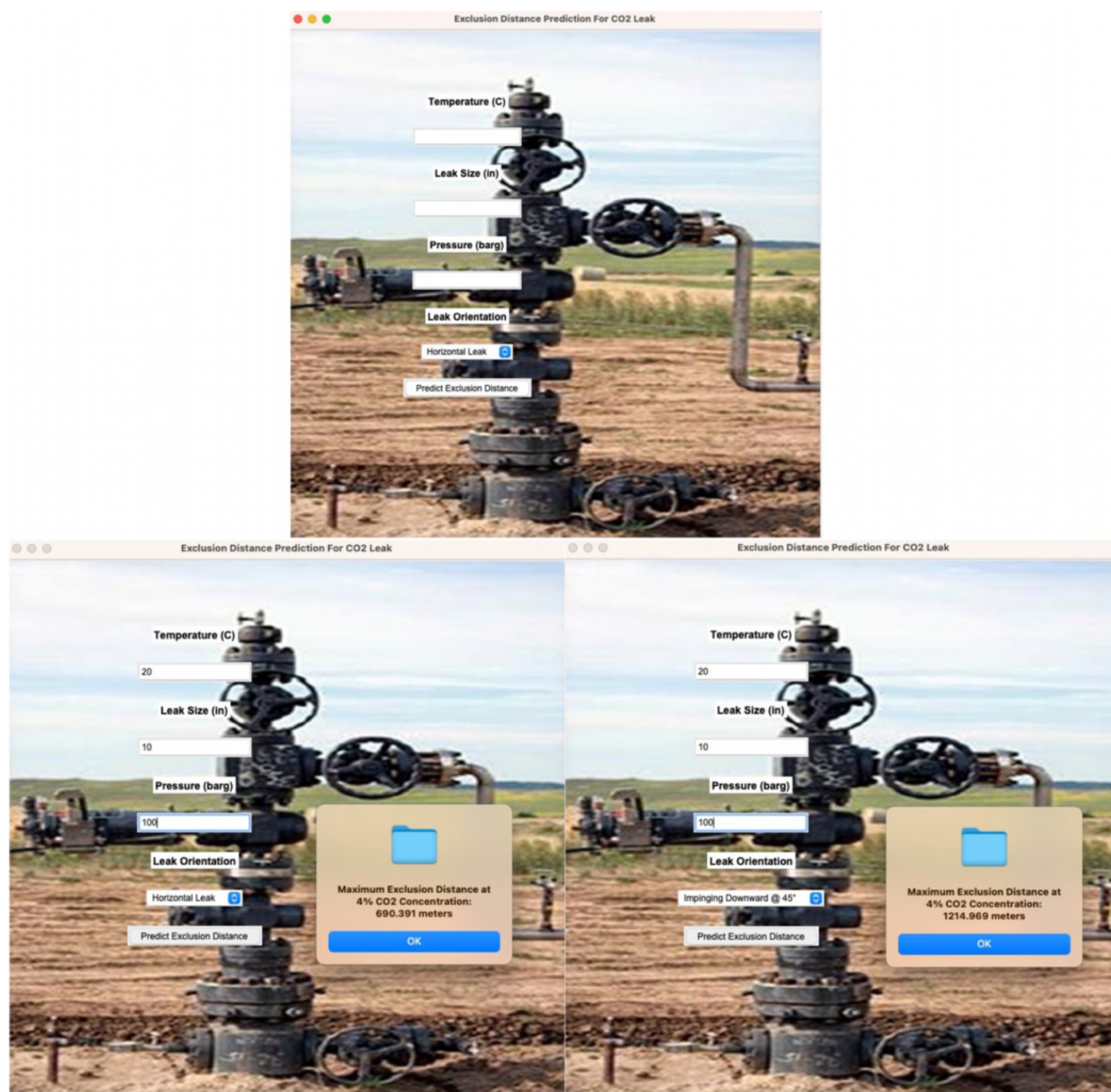


Figure 9—Screenshot of the interactive application developed using python for CO₂ leak exclusion distance prediction.

Despite the promising performance of the ANN-based dispersion prediction model, several limitations must be acknowledged. First, the model is fundamentally reliant on the quality and scope of the PHAST simulation data as any biases or errors inherent in the simulation outputs may propagate through the machine learning model. The reliability of the predicted CO₂ dispersion results is underpinned by the validation of the PHAST simulation model against real-world experiments. PHAST has been tested extensively against large-scale CO₂ release trials, including the COSHER joint industry project and the National Grid COOLTRANS experiments, which showed good agreement between PHAST predictions and observed concentration fields (Ahmad et al., 2015; Witlox et al., 2014). Nevertheless, as part of our future work, we are also pursuing

conducting field validation. This ongoing effort will further strengthen the credibility of the proposed modeling approach.

On the other hand, error analysis reveals that while the ANN achieves very low RMSE and near-perfect R^2 values, these metrics reflect performance on a controlled dataset. The potential for overfitting, despite rigorous cross-validation, remains, particularly when the operational parameter space is expanded beyond the ranges used for training. Future work should explore sensitivity analyses, incorporate broader datasets, and potentially integrate real-world measurement data to further validate the models.

Conclusions

This study extends the capabilities of machine learning-based predictive modeling for CO₂ dispersion by introducing a fourth operational parameter; leak orientation, to account for the geometric effect of impinging releases directed toward the ground. Building upon our previously published work, which focused exclusively on horizontal leak configurations, we now consider both horizontal and 45° impinging downward leaks to simulate real-world accidental release scenarios more comprehensively. A total of 13,464 PHAST simulations were conducted, systematically varying operating temperature (0–100 °C), pressure (1–250 bar), leak size (1–12 inches), and leak orientation under two representative atmospheric stability classes. For each scenario, the maximum downwind distances corresponding to critical exposure thresholds; 30,000 ppm (TLV-Ceiling) and 40,000 ppm (IDLH), were extracted to construct a high-resolution dataset.

Several machine learning models (KNN, SVR, DT, ANN, and MPR) were developed and rigorously compared using an 80/20 train-test split and 5-fold cross-validation. The ANN model demonstrated superior performance, achieving an RMSE below 10 m and an R^2 approaching 0.999 on the test data, thereby capturing the complex nonlinear dependencies governing CO₂ dispersion. The high predictive accuracy and rapid computational performance of the ANN model offer significant benefits for real-time emergency response. The results demonstrate that the inclusion of leak orientation, particularly impinging leaks, markedly increases predicted dispersion distances, reflecting the enhanced near-ground accumulation of dense CO₂ clouds. These findings have direct implications for risk management and emergency preparedness, emphasizing the need to account for release geometry in hazard zone mapping. By integrating high-fidelity physical simulations with robust machine learning techniques, this study delivers a scalable and accurate tool for predicting toxic gas dispersion in industrial settings. The model's computational efficiency and predictive reliability make it suitable for real-time applications, such as decision support systems and emergency response tools. Furthermore, the integration of this predictive tool into an interactive application can enhance situational awareness, enabling dynamic risk assessments that are essential for mitigating the consequences of accidental CO₂ releases in industrial settings. Continued refinement of these models, alongside the incorporation of broader datasets and uncertainty quantification, will further enhance their practical utility, ultimately contributing to more effective emergency response strategies and improved overall process safety.

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